Visualization of research articles on Bio*R*xiv

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1. Introduction

BioRxiv is a free online archive that provides a distribution service for unpublished preprints in the life sciences. Due to the rapid growth of science and technology, scientists tend to post their findings immediately online before submitting them to a journal. In this project, we aim to find out the characters of articles on *BioRxiv* by finishing the following six goals:

1. **The number of pages:** Explore the relationship between the number of pages of an article and its views.
2. **Website usage:** Discover the trend of website usages, such as publication and reading from 2014.
3. **Correlation between subject areas:** Find out the overlap in the discussed topics in different subject areas.
4. **Hot topics among all articles:** Analyze possible correlations of the issues discussed in each piece.
5. **A country's academic performance:** Evaluate the academic performance (including productivity, influence, etc.) of different countries worldwide.
6. **Collaboration among institutes and countries:** Find out country and institute pairs with active cooperation.

2. Method

2.1 Data Acquisition and Database Construction

Due to the slow access to BioRxiv, our website crawling can be divided into three parts. To get a list of the articles, we first used *Octoparse* to extract the dictionaries of *BioRxiv* that contain the authors, title, subject area, and the website link of an article. Given an article's link, we used *Python* to fetch the detailed information, including abstract, institutes, and views of all the articles. Moreover, to further discover the collaboration among countries and their academic performance, we need a dataset for mapping the institutes to their countries. Therefore, we use *Octoparse* to crawl the map from https://www.4icu.org/, which contains institutes and governments in A-Z order.

After the extraction finishes and removing the duplicate articles, we obtain 92,485 pieces from **22 subject areas and 13789 institutes with their country information**. This way, we construct a *BioRxiv* article information database, with its relationship graph of three datasets shown in Fig. 1.

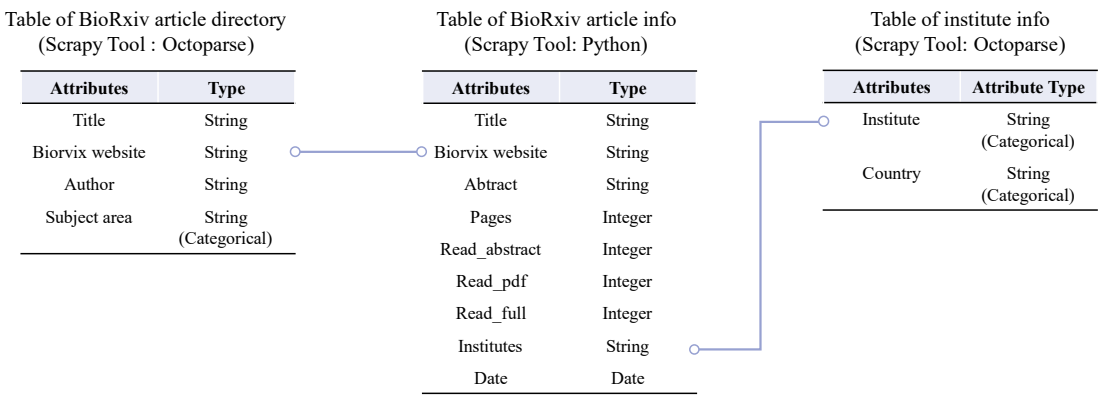


Figure 1. Graph of the relationship between three crawled datasets.

2.2 Data Cleaning and Preprocessing

2.2.1 Key Words Extraction – TF-IDF

To extract essential information from the articles' titles and abstracts, we used word frequency counts for labels of different subject areas and TF-IDF values for each abstract. Before calculating TF-IDF for each word, we first remove symbols and stop words in contents and transfer all the terms into original form to make the calculation more precise.

2.2.2 Dimension Reduction and Article Clustering

To find out the essential topics discussed on *BioRxiv*, we first applied PCA to transfer TF-IDF results into 20 dimensions. Then, we use t-SNE to find out the best 2-dimensional vector space for separating data points. Finally, we applied DBScan to cluster articles with similar topics.

2.2.3 School Match – Fuzzy Wuzzy

Each paper's institutes in the original dataset were presented using different styles and languages, making it challenging to aggregate institute information for further analysis. Here we use the Fuzzy Wuzzy package in Python to match the crawled data with a global institute list. We only chose the matched university with a similarity level greater than 90 to avoid errors.

2.3 Analysis and Visualization

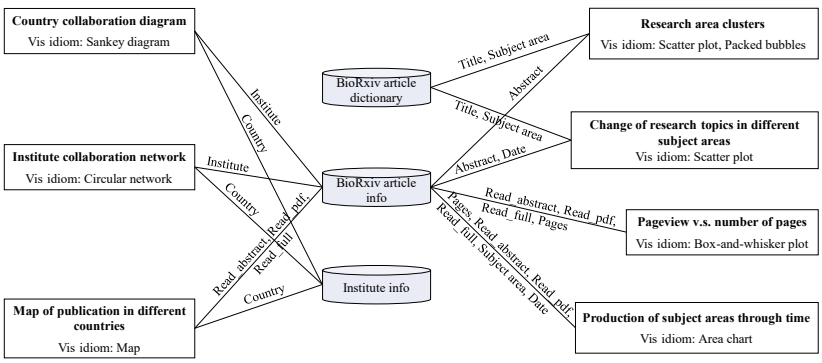


Figure 2. Data source of our seven visualizations.

2.3.1 Relationship between number of pages and views

A box-and-whisker plot is used to discover whether the number of pages an article has impacts its popularity. Page numbers are distributed into bins with a range of 30.

2.3.2 Website usage

To explore whether there is a hot research topic in different periods and the condition of attention of the publication on the website, an area chart with the color channel indicating other subject areas is adopted to present this information. The space between two lines represents each subject area's measure of interest, including the number of papers, the number of times reading abstracts, full text, or PDF. We can see the trend of website usage through time for each subject area.

2.3.3 Subject area correlation

We used the 2D-scatter plot of the dimension-reduced TF-IDF matrix and labeled each point with its subject area to find out the subject areas' correlation. By observing the coverage of each subject area in the figure and the overlapping between them, can we keep how different each subject is from research methodology and target?

2.3.4 Hot topics among all posted articles

After plotting the result of DBScan into a scatter plot, we can identify those clusters as different topics. With those clustered labels, we use the word count of titles, find the top 5 frequency words per article in each cluster, and draw word frequency per article as the size in the bubble plot. Finally, we juxtapose the cluster scatter plot and the individual result to get the final visualization dashboard.

2.3.5 Country academic performance

To discover the academic performance of each country, we visualize each country's number of papers and corresponding reading number by a map, with the illuminance of the color representing the selected academic evaluation methods (number of documents or corresponding reading amount). We can quickly locate a country with the deepest shade to have an overall understanding of countries' academic performance globally.

2.3.6 Collaboration between institutes and countries

To explore academic collaboration in Biology among the institutes and countries globally, we visualize the partnership among institutes in a circular network (plotted by *Cytoscape*) and the collaboration among countries in a Sankey diagram (plotted by *echarts* and *Python*). When representing cooperation, a network is the most suitable vis tool due to the limited connections and line size as the strength of each collaboration. This diagram uses the number of members in each institution as nodes and the number of cooperation papers between two institutions as line weight. To focus on the most active collaborators, we filter out those with a degree smaller than three and the edges with a weight smaller than 20. For the Sankey diagram, we filter out the articles that receive fewer than 1000 views and the pairs of countries that cooperate on fewer than 20 articles.

3. Results

3.1 The relationship between pages and reading amount

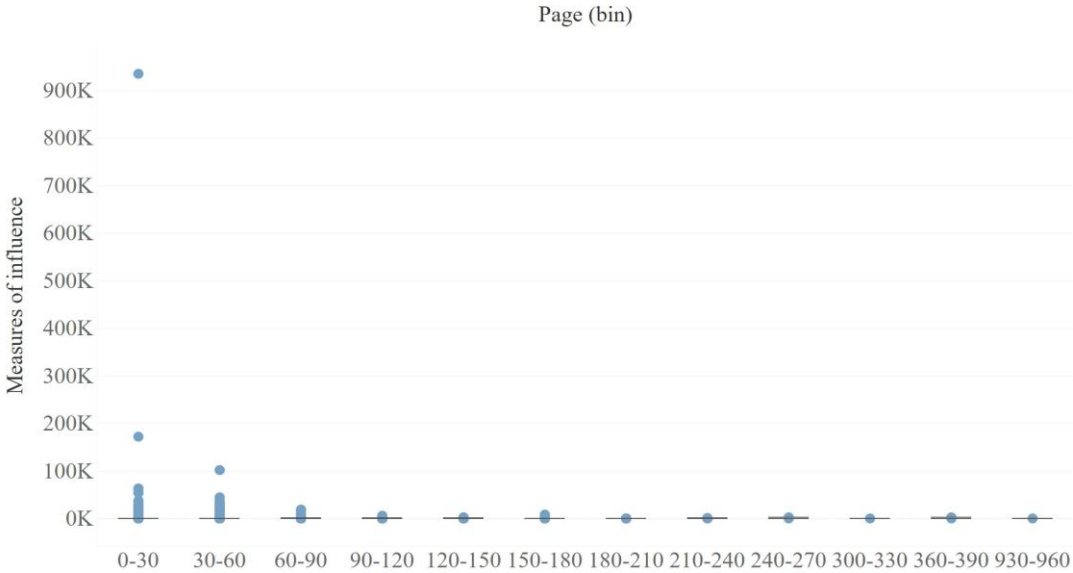


Figure 3. Page's impact on publication's reading amount.

In bins 0-30, there is an outlier that has more than 900K reading amount. As we dig into it, we find out that the website withdraws this paper. The paper claimed that 2019-CoV has similar inserts as HIV, so it caused much attention. However, the result is then proved to be spurious.

3.2 Num of publications and readings through time changes

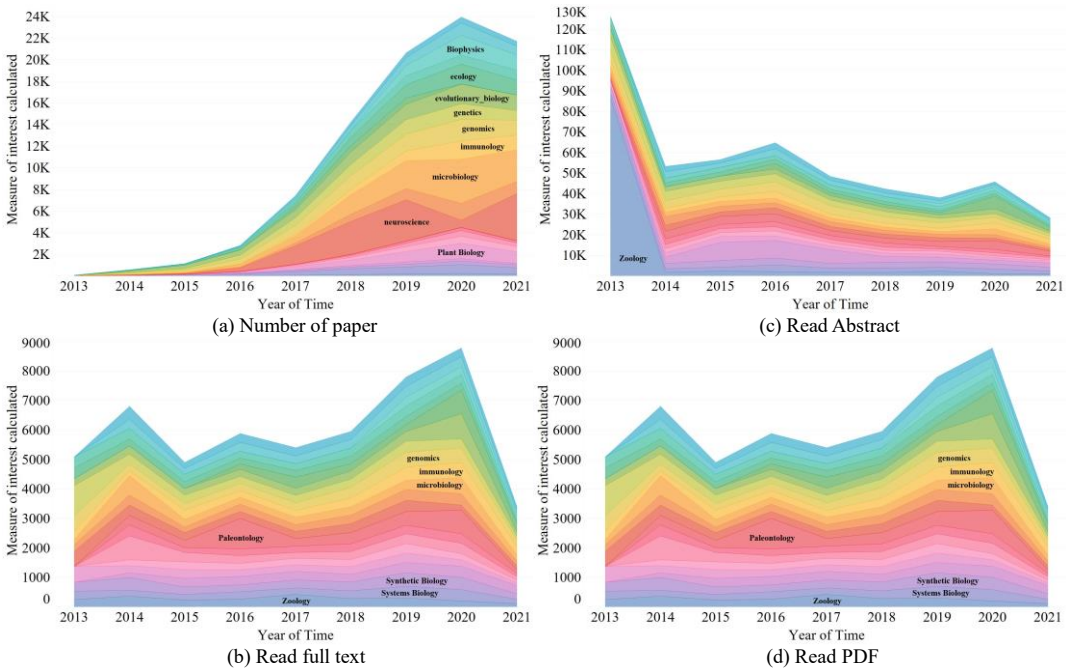


Figure.4 Website Usage chart – (a) Each subject area's number of papers over years. (b) Average number of times reading abstract. (c) Average number of times reading full text. (d) Average number of times reading PDF.

As shown in Fig. 4 (a), there is an increase in the number of publications from 2014 in all subject areas. In 2014, neuroscience is the hottest topic, and it gains attention again in 2021. In 2020, microbiology is the center of the subject area. There is a huge gap between the average abstract reading of zoology in 2013 and other subject fields. For the measure of PDF reading, the number goes down in general, which is also explainable since earlier publications have been exposed on the website longer than recent publications. Fig. 4 (b) indicates in 2020 a surge in full text reading in epidemiology and evolutionary biology, which is attributable to the COVID-19 pandemic.

3.3 The subject areas correlation

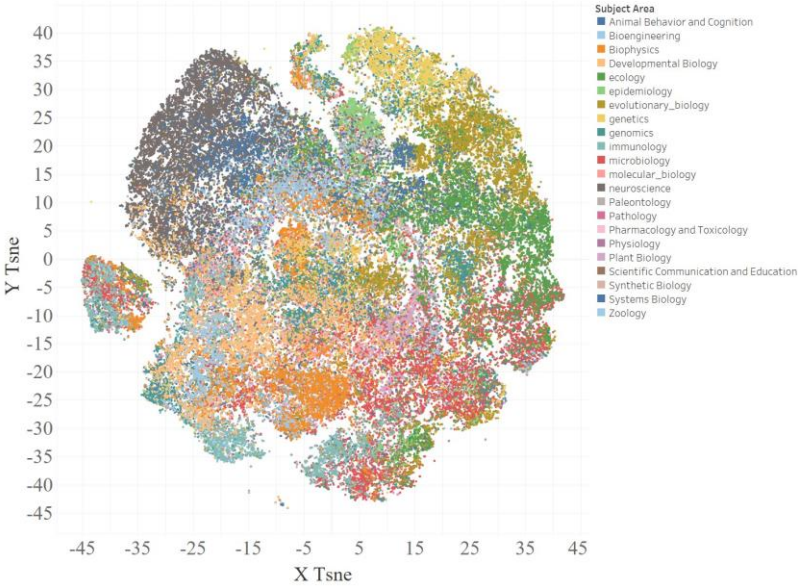


Figure 5. Subject areas correlations

Based on the scatter plot of each article, we can observe that some of the apparent overlaps among subjects. For example, bioengineering and biophysics, genetics and genomics, ecology and animal behavior, and cognition have a strong sense of overlap.

3.4 The discussed topics in Biology

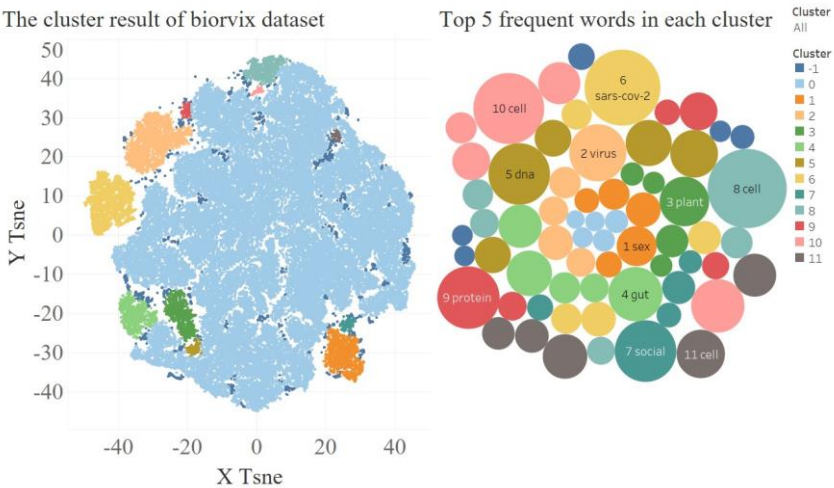
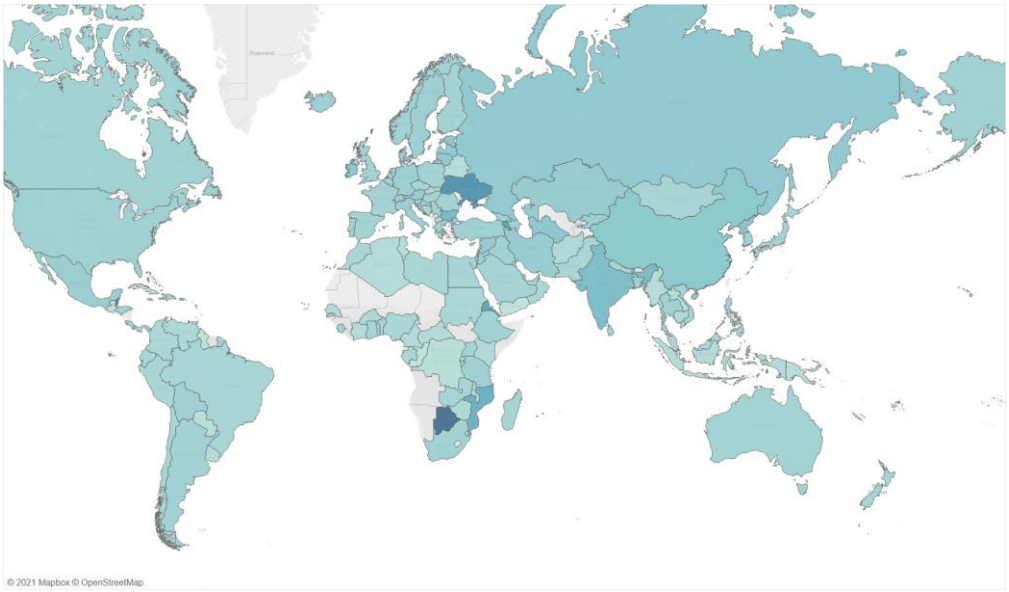


Figure 6. Cluster result of BioRxiv articles(left), Top 5 frequency words in each cluster (right, interaction)

In Fig. 6, there are 11 different topics in Biology apart from one cluster, which is hard to separate (0), and one non-distinguishable class (-1). The topic for each group is sexual behavior (1), virus (2), plants (3), guts/ Animals (4), DNA (5), COVID-19 (6), social behaviors (7), cell and immunology (8), protein (9), plants roots and cells (10) and brain (11)..

3.5 The publication of different countries

(a) Number of publications



(b) Read Full

Figure 7. Country Academic Performance – (a) Each country's number of publications. (b) Average times of reading full text.

In Fig.7, we can find that USA is the country that published the most academic paper in Biology, as shown in Fig.7(a). However, when it comes to the average time of full reading text, Botswana in Africa and Ukraine in Europe come into attention.

3.6 Collaboration between countries and institutes

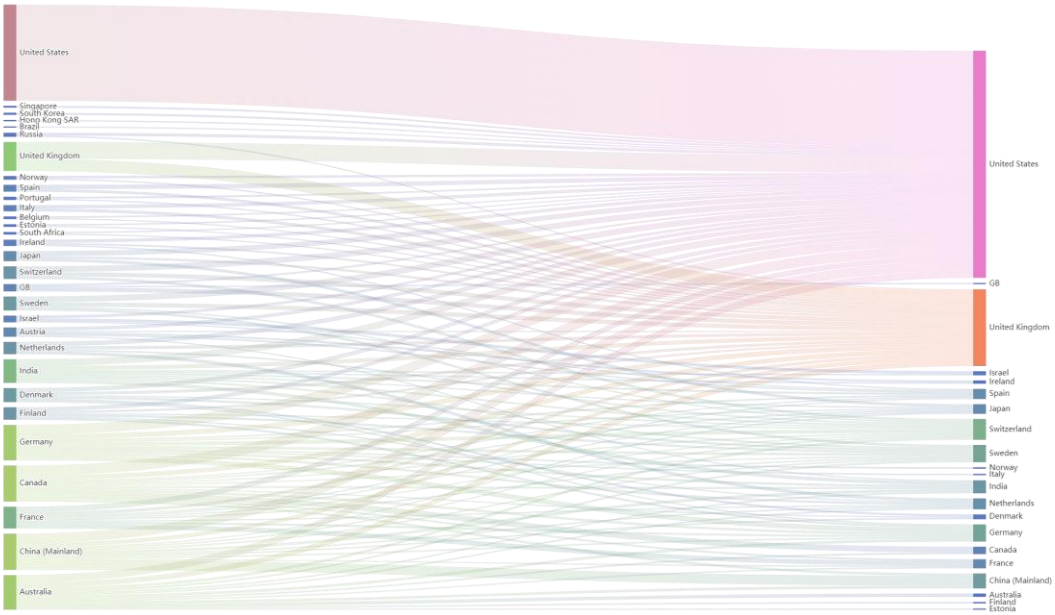


Figure 8. Country collaboration diagram. Flow width is mapped from the number of articles co-authored by researchers from the two corresponding countries. A country can have a collaborative relationship with itself.

Fig. 8 shows that the most significant volume of out-flow comes from the United States and the United Kingdom, they are the most academically collective countries regarding joint article production.

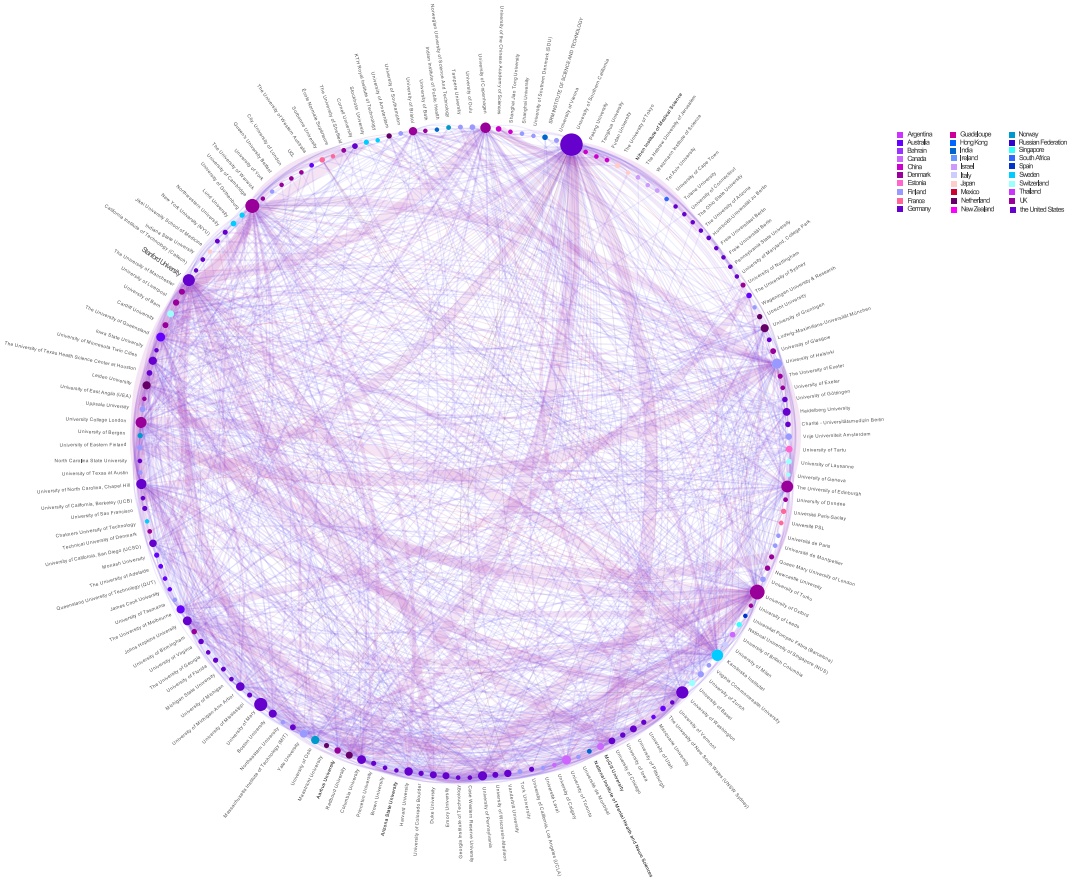


Figure 9. Institute collaboration network in circular layout. The variation in color denotes different countries and the variation in edge width denotes different times of cooperation.

In Fig. 9, we can see that USC is at the top of the list of active collaborators, followed by the University of Edinburgh, University of Oxford, Standford University, and University of Cambridge. As indicated by the most powerful strokes, the most active academic pairs are USC and UCB, Standford University and the University of Copenhagen, University of Oxford and University of Pennsylvania, etc.

4. Discussion

4.1 Color Selection

We mainly use the identity channel of color to annotate categorical attributes. We employ hue mapping for categorical data (e.g., subject areas, countries) and illumination for numerical data (e.g., page views) during color selection. For the targets related to subject areas, as shown in Fig. 4 and Fig. 5, we use "rainbow," a categorical colormap to separate the 22 subject areas. Such a choice is practical since its colors make clusters both recognizable and visually attractive. However, categorical colormaps are limited in their lack of options of both visually harmonious and distinguishable colors. As a result, when the number of attributes is large, it is hard to find suitable colors to represent all the categories. For example, Fig. 9 has many

countries, and the goal of finding compatible colors makes it harder to find an excellent categorical colormap. To visualize numerical data, as shown in Fig. 7, we adjust the illumination of blue-green color to make the countries with the most publicized articles more observable.

4.2 Interaction

Due to the substantial attributes in our dataset (multiple subject areas, institutes, and abstracts) and our versatile evaluation methods, all the figures we present are interactive. The interaction of our sculptures can be discussed in three aspects: tooltip motion for detailed information, filtering for viewing different attributes values, and the selection of other evaluation methods. In Fig. 5 and Fig. 6, the tooltips with article titles pop up when a mouse hovers over the button, which helps users better understand how the points in scatter dots are placed and why the clusters are labeled by the algorithm *DBScan* make sense.

Moreover, in Fig. 6 (right), 7, and 8, users can move their mouse to explore the exact values of each block, circle, or edge, which makes up the drawback of the inaccuracy of visualizations. As for the interaction of filtering of the subject areas, as shown in Fig. 5 and 6, it separates one subject area/cluster from others by changing the saturation of other collections. This effect both avoids messy colors and helps emphasize the sets of interest. Lastly, the different evaluation methods in Fig. 4 interact better than a static plot matrix since we need not compare them.

4.3 Size

Data visualization through size is mainly shown in Fig. 6 (right), Fig. 8, and Fig. 9, all of which have many points. Compared to color, size is easier to read at first glance. The width of lines in Fig. 8 and Fig. 9 offers users a clear idea of the most significant cooperation immediately. Moreover, the size of bubbles in Fig. 6 encodes the texts on bubbles. Only those words with the top 2 and top 3 frequency will automatically show, and the others will only show in text blocks when a mouse hovers on them.

5. Conclusion

In conclusion, the visualizations of our crawled *BioRxiv* dataset complete the tasks we set up. With the increasing number of professors willing to post their publications in *BioRxiv*, can we discover more tendencies in biological research interest's changes clearly and fluently. It can also improve the interdisciplinary subjects' growth through the research target overlaps. Finally, with more information institution cooperation be discovered and visualized, we can find the top labs professed in specific tasks efficiently.

Appendix Visualization Links

Figure	Link
Figure 3,4,7	https://public.tableau.com/views/STATS401-groupproject1-dashbard/Dashboard1?:language=zh-CN&publish=yes&:display_count=n&:origin=viz_share_link
Figure 5	https://public.tableau.com/views/STATS401-groupproject1-subjectareacorrelation/Correlation?:language=zh-CN&:retry=yes&:display_count=n&:origin=viz_share_link
Figure 6	https://public.tableau.com/views/STATS401-groupproject1-clusterresult/1?:language=zh-CN&:retry=yes&:display_count=n&:origin=viz_share_link
Figure 9	https://www.ndexbio.org/#/network/f5c100f0-135f-11ec-9e8e-0ac135e8bacf?accesskey=7cf7c3a53eab8f2bea126dfbc2f2283a64eacb46b5b67848e86730cebb8ff43c